



Engineering Gameful Systems: Insights from the 9th International GAS Workshop at ICSE 2025

Antonio Bucchiarone
Università degli Studi dell'Aquila
L'Aquila, Italy
antonio.bucchiarone@univaq.it

Beatriz Marín
Universitat Politècnica de València
Valencia, Spain
bmarin@dsic.upv.es

Kendra M. L. Cooper
Independent Scholar
Kelowna, Canada
kendra.m.cooper@gmail.com

Jeremy Bradbury
Ontario Tech University
Canada
jeremy.bradbury@ontariotechu.ca

Carlos Cetina
Universitat Politècnica de València
Valencia, Spain
cetina@upv.es

Mathieu Nayrolles
Ubisoft Entertainment
Canada
mathieu.nayrolles@gmail.com

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As gameful systems, spanning entertainment games, serious games, and gamified applications, continue to grow in complexity and societal impact, there is an urgent need for structured, interdisciplinary approaches to their engineering. This paper presents an overview and reflection on the 9th International Workshop on Games and Software Engineering (GAS 2025), held in conjunction with the IEEE/ACM International Conference on Software Engineering (ICSE 2025). The workshop brought together researchers and practitioners from software engineering and game development to address the methodological, technical, and ethical challenges of engineering gameful systems. GAS 2025 featured a dynamic program that included a keynote, peer-reviewed paper presentations, and an interactive session exploring the use of AWS for Games. Contributions covered a wide range of topics including AI-driven development, playtesting techniques, procedural content generation, HCI/CHI innovations, and lifecycle processes tailored for gameful applications. The workshop fostered rich discussions that helped identify emerging research directions, practical tools and methodologies, and community-driven initiatives for ongoing collaboration. This paper synthesizes the key insights, themes, and outcomes of GAS 2025, highlighting its role in shaping a shared research agenda at the intersection of software engineering and game studies.

1. BACKGROUND AND MOTIVATION

The game community continues to expand, encompassing stakeholders with diverse interests and objectives. To meet their needs, the research and development of entertainment games, serious games (games designed with a purpose), and gamified nonentertainment applications continues to attract substantial academic and industrial attention (e.g., [2, 6, 10, 14, 15]). Entertainment games alone represent a thriving global industry, with more than two billion players and annual revenues exceeding \$120 billion. Serious games, meanwhile, form a distinct subgenre that combines the entertainment value of games with domain-specific objectives such as education, training, health, and professional development. They are now applied in diverse domains including the arts and humanities, behavioral sciences, engineering, medicine, and business. For example, serious games can support the teaching of history, reinforce technical and soft skills in software engineering, or encourage adherence to treatment and progress in physiother-

apy. Gamified applications, in contrast, often employ a narrower selection of game elements, such as avatars, badges, and leaderboards, to motivate participation in everyday activities. In business environments, they can be used to improve focus, retention, efficiency, and satisfaction [16]. More broadly, the elements of the game have proven to be effective in encouraging user participation in areas such as sustainable transportation, the adoption of healthy lifestyles, and civic participation.

1.1 Relevance to Software Engineering

The engineering of gameful systems poses unique challenges for software engineering due to their demanding quality-of-service (QoS) requirements, ranging from accessibility and performance to scalability, usability, and security. Unlike conventional applications, gameful systems must also deliver engaging, fun, and meaningful experiences by integrating game aesthetics, mechanics, and dynamics within a player-centric design philosophy. These multifaceted demands underline the interdisciplinary complexity of designing gameful experiences.

Researchers have responded to this complexity by addressing key intersections between software engineering and gameful systems, producing results reported in leading conferences and journals. Tailored approaches have emerged in areas such as requirements engineering [3], architecture design [9], and testing [15, 19]. In the area of education, games have been designed to teach formal specifications [13] and mutation testing. Emerging research directions include the use of artificial intelligence, automation [1], and data-driven methods such as analytics and visualization [11, 7].

1.2 GAS 2025 Aims

Building on this context, the main goals of GAS 2025 are to:

- Bring together the greater communities of Software Engineering and Game Engineering in an interactive program that encourages discussion and scholarly debate from interdisciplinary perspectives.
- Identify and explore emerging opportunities, research challenges, costs, and benefits for entertainment games, serious games, and the gamification of traditional (non-games) applications and activities.

- Examine the application of SE methods and techniques to games and game-based applications, revealing novel opportunities and challenges.
- Generate a new research agenda, identify topics of interest for this community, and consider how future workshops and social media groups can explore these topics.

1.3 Topics of Interest

GAS 2025 welcomes submissions that address topics at the intersection of software engineering and entertainment games, serious games, and gamification. To highlight emerging trends and focus areas, the topics are organized into five thematic clusters:

Core Engineering Activities. Requirements engineering, architecture, design, testing, and playtesting tailored for gameful systems; lifecycle processes (e.g., Agile, DevOps); configuration and build management; and adherence to standards and ethics.

Artificial Intelligence and Automation. Applications of AI in development and gameplay; adaptive systems; procedural content generation; machine learning for bug detection and triage; and the use of intelligent agents in testing and evaluation.

Data, Analytics, and Evaluation. Large-scale gameplay data monitoring and visualization; metrics and biometrics for player behavior and system performance; qualitative and quantitative impact assessment; and reuse of data for product lines.

Immersive and Emerging Technologies. Virtual, augmented and mixed reality; games in the metaverse and other virtual environments; wearable computing; and new approaches to interaction design.

Education, Tools, and Infrastructure. Game-based education and training (including software engineering education); competitions such as game jams; domain-specific languages and formal methods; and infrastructure support for scalable game development.

Through these goals and topics, GAS 2025 is positioned as a dedicated forum to advance the state of the art in the engineering of gameful systems. By fostering interdisciplinary dialogue, the workshop seeks to chart new research directions, encourage innovative practices, and strengthen collaboration between the software engineering and game engineering communities.

2. WORKSHOP CHAIRS

The 9th edition of the GAS 2025 workshop has been organized by:

- Jeremy Bradbury, Ontario Tech University, Canada
- Antonio Bucchiarone, Università degli Studi dell'Aquila, Italy
- Carlos Cetina, Universitat Politècnica de València, Spain
- Kendra M. L. Cooper, Independent Scholar, Canada
- Beatriz Marín, Universitat Politècnica de València, Spain
- Mathieu Nayrolles, Ubisoft Entertainment, Canada

3. RESEARCH CONTRIBUTIONS AT GAS 2025

The GAS 2025 workshop received 18 submissions, of which 6 were accepted after a rigorous peer-review process. Each paper contributes a unique perspective at the intersection of games and software engineering, ranging from architectural patterns and testing methodologies to AI-driven animation, ontology education, and security. In the following, we summarize the accepted contributions.

3.1 A Mapping Study of the Entity Component System Pattern

This paper [17] presents the first systematic mapping study of the Entity Component System (ECS) pattern, a data-oriented architectural approach increasingly adopted in real-time applications and game engines. From 82 initial results, the authors identified 25 relevant studies, which they analyzed to map research trends, benefits, and drawbacks of ECS. The results show that ECS offers modularity, cache efficiency, and runtime flexibility, which are particularly suited to the demands of live-service games. However, drawbacks such as the complexity of debugging and limited academic understanding persist. The work provides a foundation for future comparative studies between ECS and other architectures, calling for deeper empirical research on the long-term impact of ECS on maintainability and scalability.

3.2 Generative AI for Facial Expressions in 3D Game Characters: A Retrieval-Augmented Approach

This contribution [4] explores how Retrieval-Augmented Generation (RAG) can be used to produce dynamic and expressive facial animations for 3D characters. By combining large language models with retrieval mechanisms, the authors propose a system that generates contextually appropriate expressions based on Facial Action Coding System (FACS) units. Their prototype demonstrates how RAG-driven NPC animations can improve immersion by enabling adaptive, emotionally responsive interactions. The paper situates this work within ongoing debates about procedural versus AI-driven animation, highlighting RAG's promise to advance personalization and realism in interactive storytelling.

3.3 A Mapping of Recording-based Game Test Automation Tools

Focusing on automated testing, this paper [8] maps recording-based tools that capture and replay user interactions in games. Through a systematic review of 20 tools identified from industry blogs and documentation, the authors classify them into general-purpose and game-specific solutions. General-purpose tools (e.g., Selenium, Appium) show adaptability but are limited to UI and visual verification, while game-specific tools integrate more deeply with engines, enabling functional and gameplay testing. The study highlights the gaps in supporting multiplayer stability and large-scale automated playtesting. Its contribution lies in providing a taxonomy of tools, their supported features, and a roadmap for designing more reliable, scalable testing infrastructures for games.

3.4 Towards a Framework for Exploratory Testing in Video Games

This paper [5] proposes a structured framework for exploratory testing (ET) tailored to the unique needs of video games. Building on traditional ET literature and previous research, the framework defines nine strategies (e.g., Newbie Journey, Golden Path, Completionist) with associated guidelines and sequencing. In an empirical evaluation on an indie platformer, the framework uncovered 80 bugs, half of them previously unknown, demonstrating its

effectiveness. The article emphasizes the importance of human creativity and domain knowledge in complementing automated approaches, positioning exploratory testing as a critical practice to ensure the quality for game play and user experience.

3.5 Type Token: A Competitive, Collaborative Modelling Game for Ontology Education and Development

Addressing challenges in ontology education, this paper [18] introduces *Type Token*, a multiplayer game designed to teach and reinforce foundational ontological concepts based on the Unified Foundational Ontology (UFO). Through collaborative and competitive modeling with tangible cards, players engage with concepts such as type-token distinctions and types of types. Playtests at conferences and workshops showed strong participation, group discussion, and evidence of activated ontological thinking. Beyond its educational utility, the paper highlights the potential of serious games to support collaborative ontology development through dialogue and rule-based interaction, pointing toward new participatory practices in knowledge engineering.

3.6 Preventing Client-Side Exploits in Games with Capability Architectures

This paper [12] addresses the persistent issue of cheating in online games by using capability architectures to protect client-side data. Unlike invasive kernel-level anti-cheat mechanisms, capability architectures compartmentalize sensitive memory structures with tagged pointers, preventing exploits such as wall-hacks and aimbots while preserving player privacy. Using a modified version of the open-source shooter *OpenArena*, the authors demonstrate a prototype that integrates machine learning detection within this architecture. Their approach represents a promising balance between effectiveness, efficiency, and ethical considerations, offering a path toward hardware-supported, privacy-friendly anti-cheat systems.

3.7 Synthesis of Contributions

Together, these six papers illustrate the rich interdisciplinarity that defines GAS. They collectively address **testing and quality assurance**, the use of **AI and automation**, **architectural innovation**, and **education and engagement**. By spanning foundational patterns, practical tools, and conceptual frameworks, the accepted papers not only advance software engineering research, but also reaffirm the mission of GAS to bridge communities and foster dialogue around the engineering of gameful systems.

4. ADVANCES AND OPPORTUNITIES IN GAME TESTING AND TRIAGE

The GAS 2025 keynote was delivered by **Sarra Habchi**¹, a leading voice in software quality and testing at Ubisoft La Forge. In her talk, she addressed one of the most pressing and persistent challenges in game development: ensuring software quality in the face of escalating complexity.

Habchi began by framing the scale of the modern video game industry, which now exceeds \$200 billion in annual revenue. Far beyond simple entertainment artifacts, today's games are *highly interconnected software ecosystems*, integrating rendering engines, artificial intelligence, physics simulations, network systems, and more. As games grow in both scale and system complexity, so does the burden of verifying their behavior across countless user interactions and configurations.

¹<https://habchisarra.github.io/>

Despite advances in tools and frameworks, **game testing remains largely manual**. Habchi illustrated how the interactive, stochastic, and non-linear nature of gameplay makes automation particularly difficult to implement. Games present a fundamentally different testing challenge from traditional applications, as test oracles are often ambiguous, expected behaviors are hard to specify, and emergent behavior can vary widely across playthroughs. Furthermore, the massive number of assets and game states makes full coverage virtually impossible.

Another significant pain point highlighted was **issue triage**, the process of prioritizing, clustering and assigning thousands of bugs and user-reported issues that arise during testing and post-release. Developers are often overwhelmed by the volume of data, leading to delays in resolution and potential quality regressions.

However, Habchi's keynote struck an optimistic tone. Drawing on both academic research and Ubisoft's internal R&D efforts, she showcased *promising automation strategies* that are beginning to alleviate these pressures. These include machine learning models to cluster similar bug reports, heuristic approaches to identify critical regressions early, and techniques to embed testability into the game architecture from the ground up. She also explored how **continuous integration pipelines** are being adapted to accommodate the dynamic content and systems of modern games.

Importantly, she emphasized that while **automation is gaining ground**, it is not a one-size-fits-all solution. Many approaches remain brittle, domain-specific, or limited by the lack of formal specifications for game behavior. Habchi closed by urging the software engineering research community to engage more deeply with the unique challenges of game testing, particularly in areas such as oracle generation, player-centric quality metrics, test reuse, and adaptive test planning.

Her keynote was not only a survey of the current landscape, but also a *call to action*, highlighting how cross-disciplinary research in software engineering, artificial intelligence, and HCI can help transform quality assurance in one of the most demanding and innovative domains of software development.

5. INTERACTIVE ACTIVITY: GENERATIVE AI IN GAME DEVELOPMENT

A standout feature of GAS 2025 was the interactive session hosted by **Amazon Web Services (AWS) for Games**, which explored how generative AI is reshaping the game development pipeline, from content creation and design to live operations and publishing. Led by **Arya Subramanyam**² and **Andrei Muratov**³, the session provided an in-depth look at both the technological capabilities and the emerging research challenges at the intersection of AI and games.

Andrei Muratov opened the session with a broad perspective on how generative AI is being adopted across game studios. His overview positioned AI not only as a development tool but as an enabler of adaptive, player-driven experiences. He emphasized that generative AI now plays a role in everything from asset creation and NPC behavior design to publishing workflows and community engagement. This highlights a shift from deterministic pipelines to *probabilistic, learning-based game engineering*, raising new questions about validation, stability, and user perception.

²<https://www.linkedin.com/in/aryasub/>

³<https://www.linkedin.com/in/andrei-muratov/>

Arya Subramanyam then delivered a focused and practical walk-through of AWS's machine learning ecosystem, with particular emphasis on **Amazon Bedrock** and its integration of large language models (LLMs), agents, and knowledge bases. Through live demonstrations and use-case walkthroughs, she illustrated how techniques such as prompt engineering, retrieval-augmented generation (RAG), and prompt chaining can be used to tailor generative systems to specific in-game contexts.

Rather than merely showcasing capabilities, the session critically engaged with real-world applications, including AI-driven architecture design in *Saltwater Games' "Resurgence"* and analytics-heavy decision systems in *HudStats*. These case studies grounded the conversation in practical implementation but also revealed limitations in current tooling, particularly around data governance, content alignment with game logic, and system interpretability.

The session ended with an open discussion on unresolved research challenges. The key among them were: the **contextual grounding** of LLM within dynamic, rule-based environments; methods for **co-creation of humans with AI** during game design; **real-time adaptability** in low-latency, multiplayer contexts; and the **ethical implications** surrounding content moderation, player manipulation and data privacy.

In general, this session did more than present tools; it revealed a changing paradigm in game engineering. Generative AI is not just an automation layer, but a design collaborator, an operational agent, and a catalyst for *to rethink how the principles of software engineering apply to interactive, creative systems*. The research community stands at the threshold of redefining the software lifecycle for gameful systems, with AI as both a challenge and an opportunity.

6. REFLECTIONS AND EMERGING THEMES

Across the papers, keynotes, and interactive activities, a common thread emerged: The engineering of gameful systems demands solutions that balance technical rigor with human-centered considerations. Quality assurance remains a central challenge, and testing and triage are addressed through both automation and exploratory approaches. At the same time, advances in AI and architectural innovation showed how new technologies can redefine design, security, and the experience of the player. Finally, contributions to education and participation emphasized the value of game-based methods for knowledge transfer and participation.

In general, GAS 2025 reinforced its role as a forum for identifying open challenges, such as reliable testing oracles, ethical AI practices, and empirical methods for studying engagement, and for charting collaborative research directions at the intersection of software and game engineering.

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